



MatPower Converter Plugin

Konverzija MATPOWER modela i statička analiza toka snage
u pravougaonim koordinatama za dTwin

Emir Krdžalić

Problem



MATPOWER format

Mreža je opisana kroz .m case fajl — standard u analizi toka snage.



Bez MATLAB-a

dTwin okruženje nema pristup MATLAB-u, pa se .m fajl ne može direktno učitati.



Potreban direktan put

Treba pretvoriti .m fajl direktno u model koji dTwin rješava jednačine.

Cilj projekta



Konverzija

.m → .dmodl bez MATLAB-a



Rješavanje

Newton-Raphson u pravougaonim koordinatama pomoću dTwin



GUI

Jednostavno upravljanje kroz dTwin plugin



Verifikacija

Poređenje sa referentnim MATPOWER rezultatima

Matematički model — pravougaone koordinate

$$V_i = e_i + jf_i$$

Napon svake sabirnice preko realne (e) i imaginarne (f) komponente — daje isključivo polinomske jednačine, bez sinusa/kosinusa.

Gradnja Ybus matrice (π -model grane)

$$t_{pq} = a_{pq} e^{j\varphi_{pq}}, \quad y_{pq} = \frac{1}{r_{pq} + jx_{pq}}, \quad y_{pq}^{sh} = \frac{jb_{pq}}{2}$$

$$Y_{pp}^{(pq)} = \frac{y_{pq} + y_{pq}^{sh}}{|t_{pq}|^2}, \quad Y_{qq}^{(pq)} = y_{pq} + y_{pq}^{sh}, \quad Y_{pq}^{(pq)} = -\frac{y_{pq}}{t_{pq}^*}, \quad Y_{qp}^{(pq)} = -\frac{y_{pq}}{t_{pq}}$$

Jednačine snage po tipu sabirnice

$$P_i = \sum_j [G_{ij}(e_i e_j + f_i f_j) - B_{ij}(e_i f_j - f_i e_j)]$$

$$Q_i = \sum_j [G_{ij}(f_i e_j - e_i f_j) - B_{ij}(e_i e_j + f_i f_j)]$$

PQ (tip 1)

Potrošačka sabirnica

P i Q su zadani → dvije jednačine

PV (tip 2)

Generatorska, regul. napona

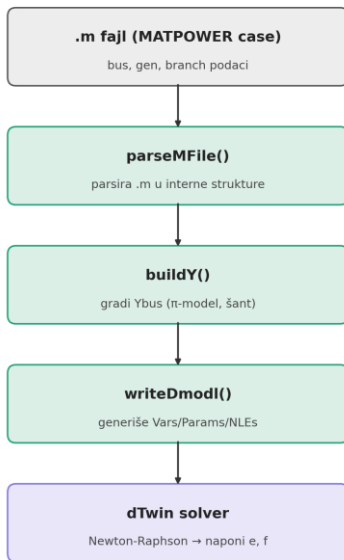
P zadano + $|V|^2$ fiksirano

Slack (tip 3)

Balansna sabirnica

e, f fiksni — ne ulaze u NLE sistem

Arhitektura i implementacija



parseMFile()

Parsira .m fajl → bus / gen / branch



buildY()

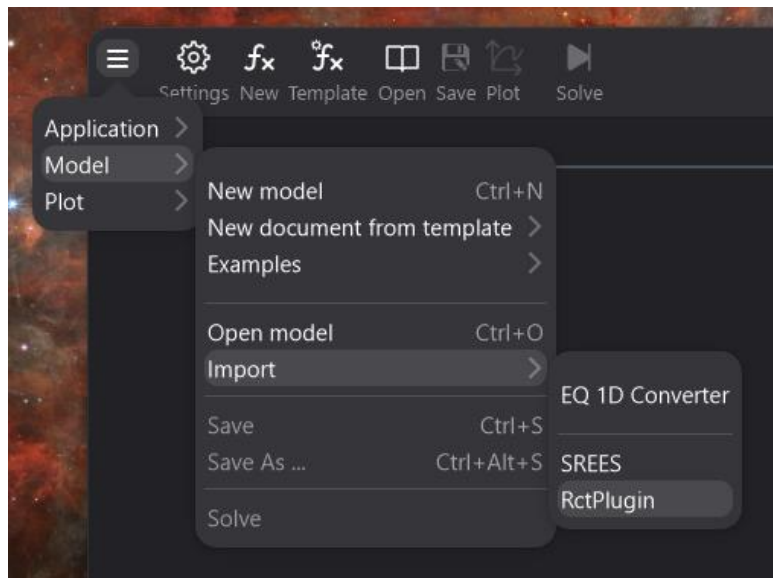
Gradi Ybus matricu (π -model)



writeDmodl()

Generiše Vars / Params / NLEs

Grafički interfejs (GUI)



Stavka „RctPlugin” u meniju Model → Import

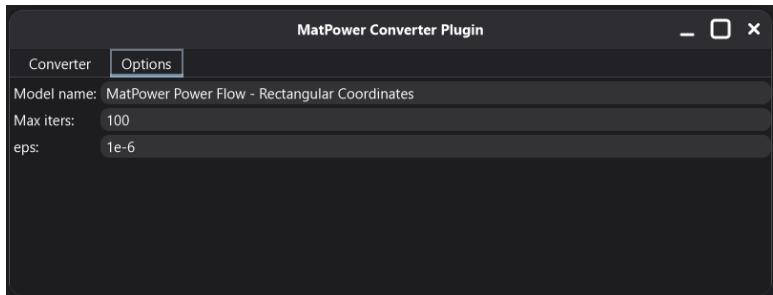
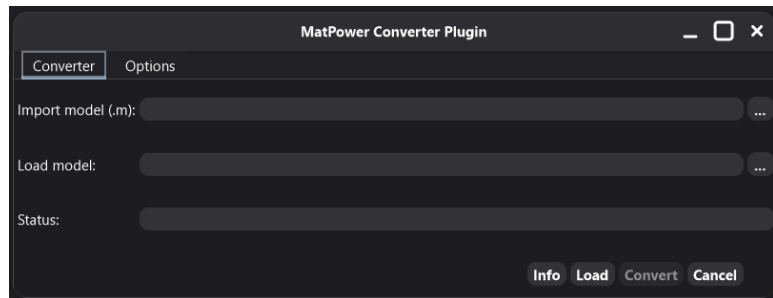
Converter tab

- Import model (.m) → automatski prelaz u Convert mod
- Save as / Load model — dinamička labela polja
- Info / Load / Convert / Cancel dugmad

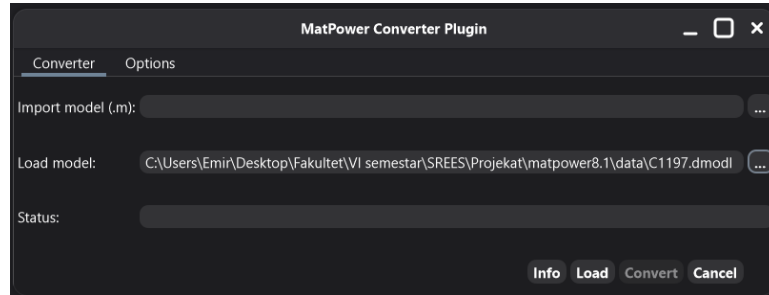
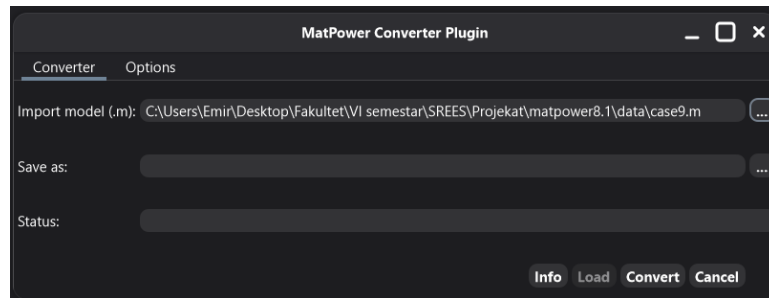
Options tab

- Model name — naziv u header .dmodl fajla
- Max iters — default 100
- eps — tolerancija konvergencije, default 1e-6

Grafički interfejs (GUI)



Converter i Options tab



Import model i Load model

Testiranje i rezultati

```
33 G / B = -1.617122473; B / B = 13.6979786
34 G 8 8 = 2.804726853; B 8 8 = -35.44561313
35 G 8 2 = 0; B 8 2 = 16
36 G 0 7 = -1.637122473; B 8 7 = 13.6979786
37 G 8 9 = -1.187604379; B 8 9 = 5.975134533
38 G 9 9 = 2.552792093; B 9 9 = -17.3382301
39 G 9 4 = -1.365187713; B 9 4 = 11.60409556
40 G 9 8 = -1.187604379; B 9 8 = 5.975134533
41 P 2 = 1.63
42 Vsp2_2 = 1.050625
43 P 3 = 0.85
44 Vsp2_3 = 1.050625
45 P 5 = -0.9; Q 5 = -0.3
46 P 7 = -1; Q 7 = -0.35
47 P 9 = -1.25; Q 9 = -0.5
48 NLE5:
49 G 2 2*(e 2 2+f 2 2) + G 2 8*(e 2 8+f 2 8) - B 2 8*(e 2 8-f 2 8) = P 2
50 e 2 2 + f 2 2 = Vsp2_2
51 G 3 3*(e 3 3+f 3 3) + G 3 6*(e 3 6+f 3 6) - B 3 6*(e 3 6-f 3 6) = P 3
52 e 3 3 + f 3 3 = Vsp2_3
53 G 4 4*(e 4 4+f 4 4) + G 4 1*(e 4 1+f 4 1) + G 4 5*(e 4 5+f 4 5) - B 4 5*(e 4 5-f 4 5) + G 4 9*(e 4 9+f 4 9)
54 - B 4 1*(e 4 1-f 4 1) + G 4 5*(e 4 5-f 4 5) - B 4 5*(e 4 5-f 4 5) + G 4 9*(e 4 9-f 4 9)
55 G 5 5*(e 5 5+f 5 5) + G 5 4*(e 5 4+f 5 4) - B 5 4*(e 5 4-f 5 4) + G 5 6*(e 5 6+f 5 6) - B 5 6*(e 5 6-f 5 6) = P 5
56 - B 5 4*(e 5 4-f 5 4) + G 5 6*(e 5 6-f 5 6) - B 5 6*(e 5 6-f 5 6) + G 5 7*(e 5 7+f 5 7) + G 5 8*(e 5 8+f 5 8)
57 - B 5 7*(e 5 7-f 5 7) + G 5 8*(e 5 8-f 5 8) - B 5 8*(e 5 8-f 5 8) + G 5 9*(e 5 9+f 5 9) - B 5 9*(e 5 9-f 5 9)
58 G 6 6*(e 6 6+f 6 6) + G 6 3*(e 6 3+f 6 3) - B 6 3*(e 6 3-f 6 3) + G 6 5*(e 6 5+f 6 5) - B 6 5*(e 6 5-f 6 5)
59 G 7 7*(e 7 7+f 7 7) + G 7 5*(e 7 5+f 7 5) - B 7 5*(e 7 5-f 7 5) + G 7 8*(e 7 8+f 7 8) - B 7 8*(e 7 8-f 7 8)
60 - B 7 5*(e 7 5-f 7 5) + G 7 8*(e 7 8-f 7 8) - B 7 8*(e 7 8-f 7 8) + G 7 9*(e 7 9+f 7 9) - B 7 9*(e 7 9-f 7 9)
61 G 8 8*(e 8 8+f 8 8) + G 8 2*(e 8 2+f 8 2) - B 8 2*(e 8 2-f 8 2) + G 8 7*(e 8 7+f 8 7) - B 8 7*(e 8 7-f 8 7)
62 - B 8 2*(e 8 2-f 8 2) + G 8 7*(e 8 7-f 8 7) - B 8 7*(e 8 7-f 8 7) + G 8 9*(e 8 9+f 8 9) - B 8 9*(e 8 9-f 8 9)
63 G 9 9*(e 9 9+f 9 9) + G 9 4*(e 9 4+f 9 4) - B 9 4*(e 9 4-f 9 4) + G 9 8*(e 9 8+f 9 8) - B 9 8*(e 9 8-f 9 8)
64 - B 9 4*(e 9 4-f 9 4) + G 9 8*(e 9 8-f 9 8) - B 9 8*(e 9 8-f 9 8) + G 9 9*(e 9 9-f 9 9) = Q 9
65 end
```

Rezultati konverzije i rješavanja za case9.m u dTwin-u

2

iteracije do konvergencije

9 / 14

sabirnica (case9 / case14)



poklapanje sa MATPOWER-om

Zaključak

- ✓ Plugin uspješno povezuje MATPOWER format sa dTwin okruženjem, bez MATLAB-a
- ✓ Pravougaone koordinate → čisto polinomski sistem NLE jednačina
- ✓ Brza i pouzdana konvergencija (2 iteracije na IEEE 9-bus i 14-bus sistemima)
- ✓ GUI omogućava korištenje bez poznavanja internog .dmodl formata

Hvala na pažnji!